Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Emotionally responsive classroom wearables for real-time student self-report and haptic co-regulation**				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (v), Next-gen Communication (v), Advanced Robotics & Manufacturing (v), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Knowledge Service			***Electrical ・ Electronics***	
	Medium Category		Small Category	Medium Category	Small Category
	Human Resource Competency Development Services		Intelligent Learning Support / Management Technologies	Electrical and Electronic Components	Sensor components
Concept of Proposed Technology	Student anxiety in STEM classrooms represents a pervasive and growing challenge that threatens mental well-being and learning performance. Current affective computing techniques, such as facial recognition, leave notable usability gaps that limit their implementation in the classroom. The proposed system of electronic wearables closes these gaps by turning the student's hand and teacher's arm into a high-bandwidth transmitter and receiver of affective context, respectively. At its core is a wireless network of electromyographic (EMG) sensors and haptic tags that merge with clothing to transmit emotional signals as rich, multimodal feedback. With an AI-assisted interface, teachers will feel their classroom as they engage through our responsiveness framework. Prototypes, preliminary clinical data, and theoretical foundations demonstrate feasibility and de-risk the proposal [1,2,3,4,5]. **Objective 1: Real-time monitoring of student anxiety.** This objective will address long-standing				

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	challenges in continuous, reliable monitoring of student emotions. A virtual squeeze ball will be developed using an EMG sensor to detect the intensity of anxiety that students feel. Construct validity will be established by comparing results from the virtual squeeze ball to gold standard survey measures. **Objective 2: AI-configured haptic feedback for teacher noticing.** PI M. Flavin will introduce a more expansive concept of a haptic interface that can be modularly and comfortably attached to existing clothing. A large-language model (LLM) will interpret plain-language instructions from the teacher to configure informative patterns of tactile input. **Objective 3: Demonstrations of closed-loop anxiety noticing wearables.** A wireless network will map information from the virtual squeeze ball as haptic feedback to the teacher. The team will lead projects to design and implement an emotionally responsive teaching framework around the use of integrated wearables. Undergraduate students with intellectual or learning disabilities and their teachers will participate in a series of mathematics courses to examine usability and impact of our system
Rationale for Joint R&D Project	A joint project is timely because emotionally responsive learning environments require **in-the-moment** visibility into student state, yet teachers cannot continuously monitor visual analytics while teaching. Haptic delivery provides low-cognitive-load, eyes-free cues that fit naturally into classroom practice, while the student-driven squeeze interaction gives students agency to signal confusion, frustration, or overload without interrupting instruction or drawing attention. Commercial translation depends on capabilities best delivered by an industrial partner: manufacturable wearable packaging (durable textiles and attachments), robust wireless performance in real classrooms, practical power/charging workflows for daily use, cost-down component choices, and scalable assembly/QA. A Korean SME partner can accelerate design-for-manufacture, reliability testing, and classroom-ready product engineering (comfort, washability, child-safe construction, simple UX), while Georgia Tech focuses on the enabling sensing-to-haptics mapping, AI-assisted configuration workflows, and validation methodology in authentic learning settings [1,2,6,7].
Key Words	1. Emotionally responsive learning 2. Student self-report (virtual squeeze) 3. Wearable EMG sensing 4. Modular haptic tags / tactile language 5. Teacher + student augmentation 6. Real-time classroom support

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]		**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives		We propose to develop and validate an emotionally responsive classroom wearable platform that: • Captures **student-driven** real-time signals using an EMG-based "virtual squeeze" interaction during STEM learning activities. • Encodes these signals into **personalized, interpretable haptic patterns** that support (a) teacher awareness for rapid response and (b) student self-regulation prompts. • Enables rapid setup and iteration through a plain-language, AI-assisted configuration workflow so teachers can define and adjust mappings without programming. • Demonstrates feasibility and early efficacy signals in classroom-relevant studies, including usability, adoption drivers, teacher responsiveness, and student engagement/affect outcomes.			
R&D Target Specifications		Success criteria will include quantitative targets across four domains: 1. **Sensing validity (student-driven signal)** • Reliable EMG detection of the "virtual squeeze" gesture across diverse students and repeated sessions • Quick calibration routines to account for placement variability and individual differences in strength • Low false-trigger rate during typical classroom movement and writing 2. **Haptic communication performance** • Discriminable, low-cognitive-load haptic patterns that can be learned quickly (a small "tactile vocabulary") • Scalable multi-tag addressing so signals can encode basic classroom semantics (who/where/urgency/type) • Stable output over time (consistent feel despite battery level changes, placement shifts, or long wear) 3. **Teacher and student augmentation** • Teacher-facing mappings that support rapid triage (who needs help, urgency, and requested support type) without looking away from instruction • Student-facing cues that support self-regulation (e.g., pacing prompts, calming cues, "try a different strategy") without social stigma • Modes for individual, small-group, and whole-class use with minimal setup overhead 4. **Usability and classroom integration** • Comfortable, safe, durable wearables suitable for K–12 classroom use (including child-safe constraints) • Wireless operation that remains robust in real classrooms (interference, many devices, varying layouts) • Practical charging, storage, and cleaning workflows for daily school operation • User acceptance signals: comfort, perceived benefit, teacher willingness to adopt, and student willingness to use repeatedly			

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R&D Plan		**Milestone 1 (Months 0–12): Core hardware + classroom-safe foundations** • Design and prototype the student EMG “virtual squeeze” wearable with robust signal acquisition and basic on-device gesture detection. • Build first-generation modular haptic tags and attachment approaches suitable for garments worn by teachers and students. • Define an initial tactile “vocabulary” (small set of patterns) and establish safety/comfort constraints for classroom wear. **Milestone 2 (Months 12–24): Scalable mappings + AI-assisted configuration** • Expand to multi-tag addressing and reliable operation with many users in the same space (classroom-scale wireless performance). • Develop mapping strategies that encode classroom semantics (identity/grouping, urgency, support type) with low cognitive load. • Implement a plain-language, AI-assisted configuration workflow so teachers can set up and revise mappings quickly. • Run controlled perception/usability studies to verify discriminability, learning time, and sustained comfort. **Milestone 3 (Months 24–36): Pilot feasibility validation in authentic learning contexts** • Integrate classroom-ready packaging: durable textiles/closures, child-safe construction, washability, and practical charging/storage workflows. • Conduct pilot deployments in realistic STEM learning activities to measure feasibility, teacher responsiveness, student engagement/affect, and adoption barriers. • Evaluate reliability, maintenance burden, and factors driving repeated use; iterate on design for classroom operations. • Produce design-for-manufacture recommendations and a commercialization roadmap (product configurations, cost targets, deployment models).

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Expected R&R of Research Institute (Georgia Tech)** • Core R&D: EMG-based "virtual squeeze" sensing, signal processing, and sensing-to-haptics encoding methods • Design of the tactile "language" and personalization/calibration routines for diverse students and classroom contexts • Development of the AI-assisted configuration workflow and evaluation metrics for teacher + student augmentation • Human-subject validation: study design, IRB protocols, classroom pilots, data analysis, and iteration of mappings based on results
Expected R&R of Korean SME Partner	**Expected R&R of Korean SME Partner** • Design-for-manufacture engineering: miniaturization, component selection, scalable assembly processes, and cost-down planning • Wearable product engineering: textiles/attachments, comfort, child-safe constraints, durability, washability, and repeatable placement • System integration: power/charging solutions, wireless robustness at classroom scale, device provisioning/fleet management concepts • Reliability/QA: accelerated life testing, failure-mode analysis, supply chain planning, and production test procedures • Product pathway inputs: packaging for school procurement and operations, deployment models, and early privacy/compliance positioning

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical deliverables** • An integrated prototype platform: student EMG "virtual squeeze" wearable + modular haptic tags for teachers and students + configuration workflow. • Verified haptic communication performance (discriminability, stability over time, scalable multi-tag addressing) under classroom-relevant conditions. • Demonstrated feasibility of real-time, student-driven signaling that supports both teacher awareness and student self-regulation during STEM activities. **Educational and societal impact** • A new pathway for teachers to perceive and respond to student uncertainty/anxiety in real time without relying on screens or disruptive check-ins. • Student-facing support for self-regulation and persistence during challenging tasks, reducing frustration and promoting engagement. • Reduced stigma compared with visible interventions, and improved inclusivity for students who struggle to ask for help verbally. **Commercialization relevance** • SME-led manufacturability and reliability plan (materials, processes, QA, cost targets) suitable for daily school operation. • Positioned for next-stage scale-up with district pilots and product definition (teacher-only vs teacher+student kits; classroom set vs individual add-ons; subscription vs purchase models).

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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